MIL-L-48063A(MI)
17 August 1973
SUPERSEDING
MIL-L-48063(MU)
18 April 1972

MILITARY SPECIFICATION

LAUNCHER, ROCKET, LAU 68B/A SHIPPING ASSEMBLY

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 This specification covers the technical requirements, quality assurance and reliability provisions and special requirements not covered by the drawings for the parts and assembly of an external store type rocket launcher for shipping, storing, and launching 2.75 Inch Folding Fin Aircraft Rockets (FFAR).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of initiation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition and Special Weapons, General Specification For
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria For
MIL-W-22248	Weldments, Aluminum and Aluminum Alloy
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace, and Control Systems, Procedures For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes

FSC 1340

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MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods
MIL-STD-1167	Ammunition, Data Card
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

356AS100	Launcher, Rocket LAU 68B/A Shipping Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Parts and materials. Parts and materials used in the manufacture of the rocket launcher shall conform to the applicable specifications as set forth herein and as listed on the pertinent drawings.

3.2 Design and construction.

3.2.1 The design and construction of the launcher and parts thereof shall be in accordance with drawings and this specification. The launcher shall be capable of shipping, storing and launching 2.75 Inch Folding Fin Aircraft Rockets and will be tested using rockets with the following physical characteristics: weight - 20.5 ± 0.5 pounds; length - 53 ± 2 inches; center of gravity 22 ± 2 inches from the forward end of the rocket. The launcher shall be capable of use with both high and low speed fixed wing aircraft and rotary wing aircraft. The intervalometer (see 6.3) shall be capable of being assembled into an Aircraft Rocket Launcher and controlling its rocket firing sequence under all flight and environmental conditions.

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3.2.2 Soldering. All soldering shall be in accordance with MIL-S-45743.

3.2.3 Painting limitations. Neither primer nor paint shall be applied to threads, bearing or working surfaces as indicated in drawings and specifications.

3.2.4 Welding. Production of weldment shall not proceed before Government approval of the welding procedure and of the workmanship required by MIL-W-22248, Class III.

3.2.5 Epoxy and adhesive. Epoxy and adhesive shall conform to those specifications appearing on the applicable drawings and shall be applied in accordance with recommendations of the manufacturer.

3.2.6 Spring embrittlement. None of the springs used in the fabrication of the launcher shall break when extended or distorted to the point of minimum permanent distortion.

3.3 Launcher electrical wiring. There shall be no failure of electrical wiring due to handling or environmental conditions. .

3.3.1 Grounding. All external parts of the launcher and intervalometer, when installed, shall be at ground potential regardless of selector switch position at all operating voltages. The resistance between each detent and every other detent shall be 0.1 ohm maximum.

3.3.2 Insulation resistance. The insulation resistance of the wiring harness after installation in the launcher shall be a minimum of 2 megohms at 500 volts direct current (VDC), from conductor to conductor and conductor to ground.

3.3.3 Wiring harness continuity. The continuity resistance of the wiring harness after installation in the launcher shall not exceed 0.2 ohm.

3.4 Intervalometer performance requirements, The intervalometer shall be capable of providing firing pulses to launch nineteen (19) 2.75 Inch Folding Fin Aerial Rockets. The intervalometer shall operate in accordance with all requirements when input power of 20 to 30 VDC is supplied through a 5 ohm, 25 watt resistor. The voltage may be interrupted as a result of any type of contact bounce and or switch chatter for a minimum of 5.0 milliseconds (ms) upon application of power without interfering with the proper operation of the intervalometer.

3.4.1 Single fire. The intervalometer in single fire mode shall supply a firing pulse to two Output Pins for each of the first nine applications of power.The firing pulse shall be a minimum of 1.00 ampere for 10 milliseconds (ms) minimum to each output pin within 30 ms after application of power.

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3.4.2 Ripple fire. The intervalometer in ripple mode shall provide firing pulses to launch a full complement of nineteen (19) rockets in sequence. The secquence shall be pairs of pulses for the first nine selector switch positions. The first firing pulse shall be applied to the corresponding pin within 30 ms after application of power, and the interval between pulses shall be no less than 54 ms nor more than 70 ms. Each firing pulse shall be l.OO ampere minimum per output pin with a dwell of 10 ms minimum. The overall time in ripple mode shall not exceed 700 ms from application of power to load position.

3.4.3 Simulated firing. The intervalometer shall meet all requirements of 3.4 through 3.4.2 when the rocket motor is simulated by use of a dummy load consisting of a 1.0 plus or minus (±) 10 percent (%) ohm resistor connected for 12 ± 2 ms between each firing pin and ground. The intervalometer shall function properly when tested using a 1.0 ± 10% ohm load to simulate firing a single rocket. This will insure the intervalometer will function properly when used in a launcher which fires one rocket per pulse (a 7 tube launcher) and also in a launcher which fires a pair of rockets per pulse (a 19 tube launcher).

3.4.4 Radiation hazard provision. The intervalometer shall ground all output pins when in the load and arm positions. The intervalometer shall not sequence out of the load position when power is applied. During operation, all output pins shall be grounded except those receiving a firing pulse. Grounding shall be verified by the presence of 5.0 millivolts maximum on any grounded output during application of power. The ground circuit shall ground the output with a resistance of not more than 0.1 ohm.

3.4,5 Bypass capability. The intervalometer in the ripple mode shall be capable of continuing through the entire sequence with one or more output pins short circuited to ground or open circuited. In the single step mode, the intervalometer shall, in the event of a short circuit to ground or an open circuit on output pins or pin, attempt to fire and remain in place. The next application of power will sequence the intervalometer to the next station. The intervalometer bypass cpability shall withstand short circuit current without damage.

3.4.6 “Load” circuit. The intervalometer shall have a load and arm switch. The load position shall internally ground all output circuits and prevent the intervalometer from being electrically sequenced. The internal grounding shall be accomplished by means of the ground circuit 3.4.8.

3.4.7 “Arm” circuit. The intervalometer shall have a selector arm position to provide the ground circuit to all outputs, but will also allow for electrical sequencing. The intervalometer shall be manually switched from load position to the arm position to provide positive arming of the unit.

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3.4.8 Ground Circuit. The intervalometer shall provide a grounding circuit to insure safety during loading and operation. The ground circuit shall ground all output pins when the intervalometer is in the load or arm position. As the intervalometer is advancing through its sequence, the ground circuit shall individually ground each output except those being energized. The ground circuit shall ground the output with a resistance of not more than 0.1 ohm. Grounding shall be verified by the presence of no more than 5 millivolts on any grounded output during the application of power.

3.4.9 Sequencing. The intervalometer shall sequence through the output stations as called out on applicable drawings. The output stations may or may not be loaded. The intervalometer shall continue through the sequence pattern even if power is removed and then re-applied. The time between application of power and output pulse shall not exceed 30 milliseconds.

3.4.10 Life. The intervalometer shall meet all requirements of 3.4 through 3.4.9 after a minimum of 400 cycles in single mode and 400 cycles in ripple mode.

3.4.11 Environmental. The intervalometer shall meet all of the requirements of 3.4 through 3.4.9 before or after, as required, exposure to the following environrnental conditions:

3.4.11.1 Humidity. In accordance with MIL-STD-81O, Method 507, Procedure I.

3.4.11.2 Sand and dust. In accordance with MIL-STD-81O, Method 510.

3.4.11.3 Shock test. In accordance with MIL-STD-81O, Method 516, Procedure IV, Figure 51 6-2, Amplitude A, Duration C.

3.4.11.4 Acceleration test. In accordance with MIL-STD-81O, Method 513, Procedure II, Table 513-11.

3.4.11.5 High temperature. In accordance with MIL-STD-81O, Method 501, Procedure I for 8 hours min.

3.4.11.6 Low temperature. In accordance with MIL-STD-81O, Method 502, Procedure I for 8 hours min., at minus (-) 54 degrees (°) C (-65°F).

3.4.11.7 Salt fog test. In accordance with MIL-STD-810, Method 509.

3.4.11.8 Vibration test. In accordance with MIL-STD-81O, Method 514, Procedure I, Curve J, and Procedure II, Part 3, Curve AF.

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3.4.11.9 Static Firing. The intervalometer shall be capable of being installed in a launcher, approved by the Procuring Activity, and of firing rockets in accordance with 3.4.

3.5 Launcher performance requirements. The launcher shall safely and reliably carry and launch seven (7) 2.75 Inch Folding Fin Aircraft Rockets. The launcher shall operate in accordance with all requirements when input power of 20 to 30 VDC is supplied through a 5 ohm, 25 watt resistor. The voltage may be interrupted as a result of any type of contact bounce and or switch chatter for a maximum of 5.0 milliseconds (ms) upon application of power without interfering with the proper operation of the intervalometer.

3.5.1 Launcher functional test. The launcher shall meet all the requirements of 3.4 through 3.4.9 when the rocket motor is simulated by use of a dummy load consisting of a 1.0 ± 10% ohm resistor connected between each contact and the corresponding detent for 12 ± 2 ms duration. When the intervalometer is installed in a launcher, only the first seven (7) outputs are used.

3.5.2 Loading. When assembled in the launcher the total of all irregularities in any tube must not be sufficient to prevent loading or removal of a 2.75 Inch FFAR. The force to move the rocket through the tube shall not exceed 20 pounds.

3.5.3 Preparation for firing. Contact arm must adequately contact the rocket igniter contact surface to sufficiently transmit a pulse which shall fire a rocket.

3.5,4 Detent operation. The rocket must be positively held in place in the tube by the detent assembly regardless of launcher orientation or environment. The detent shall not release a rocket at a pull force of 310 pounds minimum and 350 pounds maximum. The rocket locking portion of the detent shall fully retract to release the rocket when 20 pounds force is applied to the detent perpendicular to the axis of the tube. The detent shall electrically ground the rocket.

3.5.5 Pressure retention. The launcher shipping assembly (356AS100) shall be pressurized with air to 5.0 pounds per square inch (psi) and the leak rate shall not exceed 3.0 centimeters per second (sec).

3.5.6 Environmental. The launcher shall safely and reliably carry and launch seven (7) 2.75 Inch FFAR after subjection to the following conditions:

3.5.6.1 High temperature. The launcher without shipping covers shall be subjected to High Temperature in accordance with MIL-STD-810, Method 501, Procedure I.

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3.5.6.2 Low temperature. The launcher without shipping covers shall be subjected to low temperature in accordance with MIL-STD-810, Method 502, at -54°C (-65°F) for 24 hours min.

3.5.6.3 Shock. The loaded launcher without shipping covers shall be subjected to shock in accordance with MIL-STD-810, Method 516, Procedure I with the exception that only the vertical and longitudinal axis shall be tested. The vertical half size shock pulses shall have a peak of 10.0 g's and an 18 ms duration and the longitudinal half sine pulses shall have a peak of 13.5 g's and an 18 ms duration. The launcher shall retain all the inert rockets without damage at all "g" levels within the tolerance.

3.5.6.4 Vibration. The loaded launcher without shipping covers shall be subjected to vibration in accordance with MIL-STD-810, Method 514, Procedure I, Curve C.

3.5.6.5 Salt fog. The launcher shall be subjected to Salt Fog in accordance with MIL-STD-810, Method 509 with the exception that the test period shall be 50 hours.

3.5.6.6 Static Loading Tests. The launcher, without shipping covers, shall be subjected to the following static loads in accordance with MIL-A-8591. The launcher center section shall be suspended from a pylon simulating a bomb rack having 14" suspension. Application of test loadings shall be achieved in such a manner that the loads are equally divided among the launching tubes and are uniformly distributed over the length of each. The side (horizontal) and downward loadings shall be applied simultaneously. Loadings shall be as follows:

a. 2750 ± 20 pounds (13 g) acting downward.

b. 1800 ± 20 pounds (8.5 g) to the side (horizontal).

c. 2750 ± 20 pounds (13 g) acting rearward.

d. 2750 ± 20 pounds (13 g) acting forward.

3.5.6.7 Rough handling. The launcher, fully loaded, shall be dropped four times from 30 ± 1 inch at 45 ± 1 degree angle onto a solid concrete surface. Subsequently, the launcher shall be subjected to and pass the Pressure Retention Test.

3.5.6.8 Firing. The launcher shall be capable of igniting all functional rocket in a normal manner in both single and ripple modes. It shall be capable of firing a minimum of twenty-five (25) rounds in succession from each tube without damage.

3.6 First article tests. This specification makes provisions for First Article Tests. Requirements for submission of First Article Samples by the contractor shall be as specified in the contract.

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3.7 Workmanship. Workmanship shall be of the highest quality consistent with the requirements to produce an end item of maximum reliability. Particular attention shall be given to the following:

a. Freedom from blemishes, burrs, defects, and sharp edges. Approved defect standards may be established.

b. Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.

c. Thoroughness of soldering, welding, brazing, painting, wiring, and riveting.

d. Alignment of parts and tightness of assembly screws and bolts.

3.7.1 Riveting. Riveting operations shall be carefully performed to insure that rivets are tight and satisfactorily headed.

3.7.2 Cleaning. The launcher shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or Purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the Performance of the inspection requirements specified herein, unless disapproved by the Government. ‘The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto,

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c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or cmomercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.1.1 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a first article sample to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 10 launchers, 12 intervalometers, 5 of each sub-assembly, and 5 of each component part. When the intervalometer is being procured separately (less the launcher), 20 intervalometers shall be submitted and tested in accordance with 4.2.2b and 4.2.2c. When the launcher is being procured, less intervalometer, para. 4.2.2b and 4.2.2c shall not apply. The first article sample shall have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.1.1c.) A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A First Article Sample or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production in excess of 90 days or whenever a change occurs in manufacturing, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. Each launcher, intervalometer, subassembly and component may be examined for each drawing dimension and drawing requirement. In addition, any of all of the following examinations and tests will be performed:

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a. Ten (10) launchers as follows:

(1) All-Initial Examination (4.4.14.1).

(2) Two (2) in accordance with High Temperature , (4.4.14.2), Low Temperature, (4.4.14.3), and Static Load, (4.4.14.9).

(3) Two (2) in accordance with Vibration, (4.4.14.6), and Static Firing, (4.4.14.10).

(4) Two (2) in accordance with Flight Firing, (4.4.14.11).

(5) Two (2) in accordance with Shock, (4.4.14.5), and Static Firing, (4.4.14.10).

(6) Two (2) in accordance with Salt Fog, (4.4.14.8), and Rough Handling, (4.4.14.7).

b. Twelve (12) intervalometers as follows:

(1) ALL in accordance with Electrical Tests, (4.4.6).

(2) ALL in accordance with Simulated Firing Test, (4.4.7).

(3) ALL in accordance with Bypass Test, (4.4.8).

(4) Two (2) in accordance with Humidity, (4.4.9.1).

(5) Two (2) in accordance with Sand and Dust (4.4.9.2).

(6) Two (2) in accordanca with Shock, (4.4.9.3).

(7) Two (2) in accordance with Salt Fog, (4.4.9.4).

(8) Two (2) in accordance with Acceleration, (4.4.9.5).

(9) Two (2) in accordance with Life, (4.4.9.6).

c. Eight (8) additional intervalometers, when intervalometers are being procured separately, as follows:

(1) ALL in accordance with Electrical Tests, (4.4.6).

(2) ALL in accordance with Simulated Firing Test, (4.4.7).

(3) ALL in accordance with Bypass Test, (4.4.8).

(4) Four (4) in accordance with High Temperature, (4.4.9 .-) and Low Temperature, (4.4.9.8).

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(5) Two (2) in accordance with Vibration, (4.4.9.9) and Static Firing, (4.4.9.10).

(6) Two (2) in accordance with Flight Firing, (4.4.9.11).

d. Five (5) sets of subassemblies in accordance with applicable drawings.

Five (5) sets of component parts in accordance with applicable drawings.

4.2.3 Rejection. If any assembly or component fails to comply with any of the requirements and inspections, the First Article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new First Article sample of double sample size or portion thereof. Until a First Article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation. The term “inspection lot” as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and method, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. Lot numbering, as required, shall be in accordance with MIL-STD-1168 Each inspection lot shall bear a unique lot number.

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4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Tube, Launcher (see dwg. 356AS144 covering a detail of dwg. 356AS100).

Categories		Defects	Method of Inspection
Critical:	None defined.		
Major:		AQL 1.00 percent	
	101.	Length overall	Gage
	102.	Inside diameter max. (both ends)	Gage
	103.	Location of aft mounting hold for detent	Gage
	104.	Location of forward mounting hole for rocket stop plate	Gage
	105.	Location of middle mounting hold for contact assembly	Gage
	106.	Wall thickness	Gage

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Minor: AQL 2.5 percent

201. Length of contact assembly aperture . . . . . Gage

202. Width of contact assembly aperture . . . . . Gage

203. Length of detent assembly aperture . . . . . Gage

204. Width of detent assembly aperture . . . . . Gage

205, Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.2.2 Detent (see dwg, 356AS143 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Width of paddle	Gage
102.	Length from aft mounting hold to aft surface of locking undercut	Gage
103.	Length from top surface to top of undercut	Gage
104.	Profile of rocket locking tang (two places)	Gage
105.	Profile of rocket locking undercut (two places)	Gage
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.3 Tube, assembly (see dwg. 356AS170 covering a detail of dwg. 356AS100.

Categories	Defects	Method of Inspection
Critical:	None defined,	
Major:	AQL 1.00 percent	
101.	Missing parts or loose or improper assembly of contact	Visual
102.	Missing parts or loose or improper assembly of detent	Visual
103.	Stop missing, loose or improperly assembled	Visual
104.	Length from detent lock to contact arm	Gage
105.	Distance detent rocket locking tang projects into tube (min.)	Gage
Minor:	AQL 1.00 percent	
201.	Contact arm mobility	Visual/Mech.
202.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.4 Tube, assembly, weldment - 1 and 2. (see dwg. 356AS189 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Angle of center of aft bulkhead stop with longitudinal axis of contact arm as applicable	Visual
102.	Length of bulkhead stop	Gage
103.	Missing parts or welds.	Visual

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Minor: AQL 0.65 percent

201. Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.2.5 Tube, assembly, weldment - 3 (center tube number 7). (See dwg. 356AS189 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
101.	Length of weldments (both ends)	Gage
102.	Location of bulkhead mounting holes	Gage
103.	Missing parts or welds	isual
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.6 Hanger beam cluster (see dwg. 356AS126 covering a detail of dwg. 3556AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
1010	Pitch diameter of mounting lug holes	Gage
102.	Minor diameter of mounting lug holes	Gage
103.	Length between mounting lug holes	Gage
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.7 Insert, threaded (see dwg. 356AS102-1 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	AQL 0.65 percent	
101.	Major diameter of outside threads	Gage
102.	Pitch diameter of outside threads	Gage
103.	Minor diameter of inside threads	Gage
104.	Pitch diameter of inside threads	Gage
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.8 Tube Cluster Assembly Launcher, Rocket, LAU 68 B/A (see dwg. 356AS168 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	AQL 1.00 percent	
101.	Launcher tube radial orientation	Visual
102.	Absence of adhesive on tube outer walls	Visual
103.	Missing parts or improper assembly	Visual
104.	Hanger beam ground wires or ground screws missing	Visual
105.	Intervalometer receptacle missing or improperly assembled	Visual
106.	Strap and strap seal assembly	Visual/Mech.

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Minor: AQL 0.65 percent

201. Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.2.9 Intervalometer (see dwg. 356AS172-1 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	AQL 1.00 percent	
101.	Length from connector to standoff, max	Gage
102.	Length from standoff to base of knob flange, min	Gage
103.	Location of standoffs	Gage
104.	Location of shaft to standoff	Gage
105.	Overall height, max.	Gage
106.	Overall width, max	Gage
107.	Pitch diameter of mounting hole threads	Gage
108.	Depth of mounting hole, min	Gage
109.	Length of knob, min	Gage
110.	Diameter of knob flange, max	Gage
Minor:	AQL 1.00 percent	
201.	Outside diameter of knurled portion of knob	Gage
202.	Evidence of poor workmanship (see 3.7)	Visual

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4.3.2.10 Barrier (see dwg. 356AS108 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Outside diameter	Gage
102.	Position of mounting holes	Gage
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.11 R.F. barrier assembly (see dwg. 356AS149 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	Legibility of Warning Marking	Visual
Major:	AQL 0.40 percent	
101.	Improper assembly	Visual
Minor:	AQL 1.5 percent	
201.	Foil overlap	Visual
202.	Evidence of Poor workmanship (see 3.7)	Visual

4.3.2.12 Fairing, rear assembly (see dwg. 356AS154 covering a detail of dwg 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Inside mating diameter	Gage
102.	Radial location of locking indents (four places)	Gage

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103. Depth of locking indents (four places) . . . . . Gage

104. Length from forward end of fairing to aft edge of indent max . . . . . Gage

105. Location of fairing clip lock pin to indent . . . . . Gage

106. Missing parts or improper assembly . . . . . Visual

Minor: AQL 1.50 percent

201. Width of locking indents (four places) . . . . . Gage

202. Markings missing or illegible . . . . . Visual

203. Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.2.13 Launcher, Pocket, aircraft, LAU 68B/A (see dwg. 356AS164 covering a detail of dwg. 356AS100).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Gaps or voids in skin weld	Visual
102.	Gaps or voids in retaining ring weld	Visual
103.	Retaining ring missing, damaged, or improperly assembled	Visual
104.	Suspension lugs missing, damaged, or improperly assembled	Visual
105.	Electrical receptacles missing, damaged or improperly assembled	Visual
106.	Intervalometer missing, damaged, or improperly assembled	Visual

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107. Toggle switch missing, damaged, or improperly assembled . . . . . Visual

108. Dust caps missing, damaged, or improperly assembled . . . . . Visual

109. Barrier, electromagnetic shielding missing, damaged or improperly assembled, Visual

Minor: AQL 1.00 percent

201. Identification, warning, fairing and instruction plates missing, damaged, illegible, or unpreserved . . . . . Visual

202. Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.2.14 Launcher, rocket, LAU 68B/A shipping assembly (see dwg. 356AS100).

Categories	Defects	Method of Inspection

Critical: 100 percent

1. Safe and arm device missing, damaged, or improperly assembled . . . . . Visual

Major: AQL 1.00 percent

101. Missing parts on shipping cover assembly . . . . . Visual

102. Improper fabrication of shipping cover assembly . . . . . Visual

Minor: AQL 1.00 percent

201. Lock wire missing from mounting lugs . . . . . Visual

202. Evidence of poor workmanship (see 3.7) . . . . . Visual

4.3.3 Testing.

4.3.3.1 Spring embrittlement. Major defects: Detent (see dwg. 356AS143), Flat Spring (see dwg. 356AS113), Compression Spring (see dwg. 356AS119), Lever (see dwg. 356AS184).A sample of 10 springs shall be selected from each lot, if one or more springs break before permanent distortion, the lot shall be rejected. The test shall be conducted as specified in 4.4.1 using equipment in accordance with 4.3.4.

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Test springs shall be discarded.

4.3.3.2 Spring hardness. Major defects: Detent, flat spring, lever. A sample of 10 springs shall be selected from each lot, if one or more springs fail to comply with the requirement, the lot shall be rejected. The test shall be conducted as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3,3.3 Torque tests.

4.3.3.3.1 Insert nut torque (see dwg. 356AS153). Major defects - Each insert nut in one skin assembly per working shift shall be tested for compliance with the torque out requirement. If any insert nut fails to withstand the torque out force, the cause for failure shall be determined and the launcher skins produced since the last test shall be rejected. The test shall be conducted as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.3.2 Launcher assembly (see dwg/ 356AS164). Major defects - Threaded insert, connector assembly, nut. A sample of launcher assemblies shall be randomly selected from each lot in accordance with MIL-STD-105, Level II and an AQL of 0.40 percent.Launcher assemblies failing to comply with the requirements shall be classified as defective. The acceptance level shall apply for each defect. The test shall be conducted as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.4 Insulation resistance (see 3.3.2). Major defect - Launchers shall be tested 100% for compliance with the requirement. Launchers failing to comply with the requirement shall be classified as defective and removed from the lot. The test shall be conducted as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.5 Continuity (see 3.3.1 and 3.3.3). Major defect - Launchers shall be tested 100% for compliance with the requirements. Launchers failing to comply with the requirements shall be classified as defective and removed from the lot. The test shall be conducted as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.6 Intervalometer electrical tests.

4.3.3.6.1 Load circuit test (see 3.4.4 and 3.4.6). Critical defect - Intervalometers shall be tested 100% for compliance with the requirements. Any intervalometer failing to comply with the requirements shall be classified defective and removed from the lot. The test shall be conducted as specified in 4.4.6 using equipment in accordance with 4.3.4.

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4.3.3.6.2 Arm circuit test (see 3.4.4 and 3.4.7). Critical defect - Intervalometers shall be tested 100% for compliance with the requirements. Any intervalometer failing to comply with the requirements shall be classified defective and removed from the lot. The test shall be conducted as specified in 4.4.6.2 using equipment in accordance with 4.3.4.

4.3.3.7 Simulated firing test (see 3.4.3). Major defect - Intervalometers shall be selected from each lot as per MIL-STD-105, General Inspection Level II, and tested for proper timing and output current. If any intervalometer fails to meet all timing and output current requirements for single and ripple modes, the lot shall be rejected. The test shall be conducted as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.3.8 Intervalometer bypass test (see 3.4.5). Major defect - The intervalometers from the simulated firing test shall be tested in single and ripple modes for compliance with the bypass requirement, In any intervalometer fails to meet the requirement, the lot shall be rejected. The test shall be conducted as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.9 Intervalometer environmental tests (see 3.4.11). Major defect - When the intervalometer is being procured separately a sample of intervalometers per MIL-STD-105, General Inspection Level II shall be selected from each lot and subject to environmental tests as specified below. If any intervalometer fails any requirement, the lot is rejected.

a. Two (2) in accordance with humidity (3.4.11.1).

b. Two (2) in accordance with salt fog (3.4.11.7).

c. Two (2) in accordance with shock (3.4.11.3).

d. Two (2) in accordance with high temperature (3.4.11.5) and low temperature (3.4.11.6).

e. Two (2) in accordance with vibration (3.4.11.8) and static firing (3.4.11.9).

The test shall be conducted as specified in 4.4.9 using equipment in accordance with 4.3.4.

4.3.3.10 Additional intervalometer environmental tests for first article (see 3.4.11).

a. Two (2) in accordance with sand and dust (3.4.11.2),

b. Two (2) in accordance with acceleration (3.4.11.4).

Two (2) in accordance with high temperature (3.4.11.5) and low temperature (3.4.11.6).

d. Two (2) in accordance with life (3.4.10).

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e. Two (2) in accordance with flight firing.

These tests shall be conducted as specified in 4.4.9.

4.3.3,11 Launcher functional test (see 3.5.1). Major Defect - A sample of launchers in accordance with MIL-STD-105 Level II and AQL of O.40 percent shall be selected from each lot and tested for compliance with timing and current output requirements. Any launcher that fails a requirement shall be considered as defective and removed from the lot. Grounding of the intervalometers shall be tested in the “load” and “arm” positions. Test shall be conducted as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.12 Rocket loading (see 3.5.2). Major defect - A sample of 32 launchers from each lot shall be gaged for facility of loading a 2.75 Inch FFAR. A plug gage simulating the rocket shall be inserted in each tube If any tube does not accept this gage the lot shall be rejected. This inspection shall be conducted as specified in 4.4.11 using equipment in accordance with 4.3.4.

4.3.3.13 Detent operation (see 3.5.4).

4.3.3.13.1 Detent pull Major defect - A sample of tube assemblies per MIL-STD-105 General Inspection Level II shall be selected from each lot and tested for detent pull force with a fixture simulating a 2.75 Inch FFAR. If any tube assembly releases the fixture before the minimum pull force is reached, the lot shall be rejected. This test shall be conducted as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.13.2 Detent retraction. Critical defect - a sample of tube assemblies per MIL-STD-105 General Inspection Level II shall be selected from each lot for the detent retraction test. If any tube assembly fails the requirement of the detent locking tang not clearing the inside diameter of the tube, the lot shall be rejected. This test shall be conducted as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.14 Pressure retention (see 3.5.5). Major defect - A sample of launcher shipping assemblies in accordance with MIL-STD-105 Level II and AQL of 0.40 percent shall be selected from each lot and subjected to the pressure retention test. The test shall be conducted as specified in 4.4.13 using equipment in accordance with 4.3.4.

4.3.3.15 Launcher environmental tests (see 3.5.6). Major defects - A sample of launchers shall be selected from each lot in accordance with MIL-STD-105 special Inspection Level S-2. If any launcher fails to comply with the requirements, the lot shall be rejected. The tests shall be scheduled as follows:

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a. One (1) in accordance with high temperature and low temperature.

b. One (1) in accordance with high temperature and shock.

c. One (1) in accordance with low temperature and vibration.

d. One (1) in accordance with shock and static firing.

e. One (1) in accordance with vibration and static firing.

When 8 launchers are selected, the three additional launchers will be tested as follows:

f. One (1) in accordance with shock and vibration.

g. One (1) in accordance with shock and static firing.

h. One (1) in accordance with vibration and static firing.

Tests shall be conducted as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.3.16 Additional launcher environmental test for first article (see 3.4.11). Major defects

a. Salt fog

b. Static load

c. Rough handling

d. Flight firing

These tests shall be conducted as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment. Equipment tabulation number ET-356AS100 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contracting officer retains the approval authority on inspection equipment.

4.3.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods.

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4.4.1 Spring embrittlement. The spring shall be stretched along its longitudinal axis, in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered permanently distorted. Any spring that breaks prior to this point shall be classed defective.

4.4,2 Spring hardness. A flat section of each spring shall be polished and tested for hardness.

4.4.3 Torque tests.

4.4.3.1 Insert nut torque. The required torque out force shall be applied between each nut and the skin in the assembly selected.

4.4.3.2 Launcher assembly. The required assembly torque shall be applied to each part.

4.4.4 Insulation resistance. Insulation resistance shall be tested using a megohmmeter, applying 500 ± 50VDC for at least 5 seconds. This test will be conducted with the intervalometer and shorting pin removed. The insulation resistance of the power circuits shall be tested from each receptacle to ground. The insulation resistance of each firing circuit shall be tested from the intervalometer connector to ground. When any firing circuit is tested, every other circuit shall be shorted to the grounded launcher.

4.4.5 Continuity. Continuity of the power and firing circuits shall be tested using an ohmmeter connected to each end of the circuit under test. Continuity of each firing circuit shall be tested from the intervalometer connector to the corresponding contact. Continuity of the ground circuit shall be tested from each power receptacle to the suspension lug in the hanger beam. The shorting pin shall be reinserted and continuity to ground tested from the power pin in the intervalometer connector to the suspension lug in the hanger beam. Continuity of the grounding circuit shall be tested from each detent to the ground pin in the intervalometer connector.

4.4.6 Intervalometer electrical tests.

4.4.6.1 Load circuit test.

4.4.6.1.1 The intervalometer shall be set to the load position. Prior to the application of power, continuity shall be measured from pin N to each of the output pins using an ohmmeter. Continuity shall be 0.1 ohms max.

4.4.6.1.2 The intervalometer shall be tested for proper operation of load circuit with 30.0-0.5 volts DC supplied through a 5.0 ± 1% ohm 25 watt resistor. Upon application of power, the unit shall not sequence. A dummy load consisting of a 1.0 ± 10% ohm resistor shall reconnected between each output pin and ground. When power is applied, voltage across

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each 1.0 ± 10% ohm load shall be measured to insure less than 5.0 millivolts (mv) are present on all output pins.

4.4.6.2 Arm circuit test. The intervalometer shall be sequenced once in either the single or ripple mode at 30.0 ± 0.5 VDC supplied through a 5 ± 1% ohm 25 watt resistor for proper operation of the Arm circuit. The unit shall be set to the arm position. Upon application of power it shall sequence, the output voltage shall be monitored across a 1 ± 10% ohm load on each pin to check for 5mv maximum on each and every unfired pin.

4.4.7 Simulated firing test. The intervalometer or launcher shall be tested for proper timing and operation with power supplied through a 5.0 ± 1% ohm 25 watt resistor. Upon application of power, the voltage shall be subjected to variations similar to contact bounce or switch chatter of 5.0 ± 1.0 ms. A dummy load consisting of a 1.0 ± 10% ohm resistor shall be connected between each output pin and ground for 12.0 ± 2ms.

4.4.7.1 Single fire operation shall be tested at 20.0 ± 0.5VDC and 30.0 ± 0.5VDC input. Time shall be measured from the application of power to the leading edge of each output pulse and time shall not exceed 30 ms max. The output current shall be measured from each pin through the 1.0 ± 10% ohm load to ground and shall be 1.00 amps, min for a duration of 10 ms, min.

4.4.7.2 Ripple fire operation shall be tested at 20.0 ± 0.5 VDC and 30.0 ± 0.5 VDC input. Time shall be measured from the application of power to the leading edge of the first output. It shall not exceed 30 ms max. Time measured between the leading edge of successive outputs shall not be less than 54 ms nor more than 70 ms. The output current measured from each pin through the 1.0 ± 10% ohm load to ground shall be 1.00 amp, min. for a duration of 10 ms, min. The overall time in ripple mode shall not exceed 700 ms from application of power to load position.

4.4.8 Bypass test.

4.4.8.1 A single fire at 20.0 ± 0.5 VDC and 30.0 ± 0.5VDC with one or more parallel output pins shorted to ground or open circuited, the intervalometer shall attempt to fire and remain in place until the next application of power.

4.4.8.2 Ripple fire at 20.0 ± 0.5 VDC and 30.0 ± 0.5 VDC with one randomly selected pin shorted to ground and another randomly selected pin open circuited; the intervalometer shall complete a full sequence and meet timing requirements on all normally loaded outputs. For the purpose of this test, a short circuit to ground shall be considered 0.2 ohms or less.

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4.4.9 Intervalometer environmental tests. After completion of each of the following tests the intervalometers shall be subjected to the bypass tests as specified in 4.4.8, and the electrical tests as specified in 4.4.6.

4.4.9.1 Humidity test. The intervalometer shall be tested in accordance with MIL-STD-810, Method 507, Procedure I. At the end of cycle 5 and cycle 10, the sample shall be examined visually for damage or defects and the Simulated Firing Test shall be performed.

4.4.9.2 Sand and Dust Test. The intervalometer shall be tested in accordance with MIL-STD-810, Method 510. Immediately upon completion of the environmental conditioning, the sample shall be examined visually for damage or defects and the Simulated Firing Test shall be performed.

4.4.9.3 Shock test. The intervalometer shall be tested in accordance with MIL-STD-810, Method 516, Procedure IV, Figure 516-2, Amplitude A, Duration C. For this environmental exposure, one intervalometer shall be placed in the load position with power applied, and the other intervalometer shall be placed in the arm position with no power applied. Immediately upon completion of environmental conditioning, the samples shall be visually examined for damage or defects and the Simulated Firing Test shall be performed. Any inadvertent advance of the intervalometer or malfunction shall be cause for rejection.

4.4.9.4 Salt fog test. The intervalometer shall be tested in accordance with MIL-STD-810, Method 509. Immediately upon completion of the environmental conditioning, the sample shall be visually examined and the Simulated Firing Test shall be performed. Following a 48 hour dry period, these tests shall be repeated.

4.4.9.5 Acceleration test. The intervalometer shall be tested in accordance with MIL-STD-810, Method 513, Procedure II, Table 513-II. During each orientation in each axis, the Simulated Firing Tests shall be performed. Immediately upon completion of the environmental conditioning with the temperature maintained, the sample shall be visually examined for damage or defects and the Simulated Firing Test shall be performed.

4.4.9.6 Life test. The intervalometer shall be connected to a power source at 28 ± 0.5 VDC and functioned for the life test. The unit shall be functioned 400 complete sequence cycles in ripple mode and 400 complete sequence cycles in single mode to give a total of 800 complete cycles. The duty cycle for single mode operation shall be 1 second “on” and 5 seconds “off”. At the completion of the 800 cycles, the Simulated Firing Test shall be repeated. The sample shall also be visually examined for damage or defects.

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4.4.9.7 High temperature test. The intervalometer shall be conditioned in accordance with MIL-STD-81O, Method 501, Procedure I with the exception that the period shall be 8 hours minimum. Immediately upon completion of the environmental conditioning, with the temperature maintained, the sample shall be examined visually for damage or defects and the Simulated Firing Test shall be performed.

4.4.9.8 Low temperature test. The intervalometer shall be conditioned in accordance with MIL-STD-810, Method 502, Procedure I at -54°C (-65°F) for 8 hours minimum. Immediately upon completion of the environmental conditioning, with the temperature maintained, the sample shall be examined visually for damage or defects and the Simulated Firing Test shall be performed.

4,4.9.9 Vibration test. The intervalometer shall be tested in accordance with MIL-STD-81O, Method 514, Procedure I, Curve J and Procedure II, Part 3, Curve AF. During a resonant dwell and at the end of vibration in each axis, a Simulated Firing Test shall be performed on the intervalometer. Immediately upon completion of the environmental conditioning the sample shall be visually examined for damage or defects.

4.4.9.10 Static firing test. The intervalometer shall be installed in a launcher. The loaded launcher (warheads and fuzes shall be inert) shall be suspended from a bomb rack or simulated bomb rack having 14 inch suspension provisions and test fired. The potential at the launcher connector shall be 20.0 ± 0.5 volts DC. There shall be a 5 ohm limiting resistor in series with the launcher intervalometer. The firing sequence shall consist of ripple firing the loaded launcher, reloading and single fire. The intervalometer shall fire all functional rockets with a time to the first firing pulse of 30 ms max and intervals of not less than 54 ms nor more than 70 ms thereafter. In the single fire mode each pulse shall appear in 30 ms from application of power. Upon completion of firing the sample shall be visually examined for damage or defects and the Simulated Firing Test shall be performed.

4.4.9.11 Flight firing test. The intervalometer shall be installed in a launcher. The launcher shall be loaded with 2.75 Inch FFAR and installed on a bomb rack. The launcher shall be fired in flight to determine conformance of intervalometer with requirements and ability to withstand air loads and vibration imposed by aircraft. The test shall be to fire a fully loaded launcher in ripple mode, reload and fire in single mode. All rockets shall fire with proper timing.

4.4.10 Launcher functional test. The launcher shall be tested for proper timing, output current and grounding at 20 ± .5VDC supplied through a 5 ohm + 1% 25 watt resistor in series with one of the power connectors. Upon application of power, the voltage shall be subjected to variations similar to contact bounce or switch chatter for 5.0 ± 2.0 ms. A dummy load consisting of a 1.0 ± 10% ohm resistor shall be connected between each contact and the corresponding detent for 12 ± ms. The following values shall be measured:

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4.4.10.1 Single fire. Time to each pulse after application of power, output current of each pulse and dwell time for each pulse.

4.4.10.2 Ripple fire. Time to first pulse after application of power, output current of each pulse, dwell time for each pulse and time between the leading edges of each pulse.

4.4.10.3 Grounding. Apply power when the intervalometer is in the load position and monitor voltage across each 1.0 ohm load for 5.0 mv max.

4.4.11 Rocket loading. A plug gage simulating a rocket shall be inserted from the forward end of the tube to the detent. The plug gage shall be 2.800 + .002 or - .000 inches in diameter for a length of 42.0 inches and a straightness of .003 inches. The gage may be of continuous diameter through the length or a series of gaging surfaces spaced no more than 6.0 inches apart. Not more than 20 pounds force will be used to insert or remove the gage.

4.4,12 Detent operation.

4.4.12.1 Detent pull. A fixture simulating the 2.75 Inch FFAR with a groove representing the locking groove on the nozzle and fin assembly shall be inserted into the tube and locked into place by the detent. Pull force shall not exceed 350 pounds to prevent damaging the tube assembly.

4.4.12.2 Detent retraction. The detent shall be lifted by a lever assembly designed to exert 20 pounds force vertically on the exposed end of the detent. When the force is applied the inspector shall mechanically or visually verify that the rocket locking portion of the detent is completely lifted out of the inside diameter of the tube.

4.4.13 Pressure retention. The launcher shall be pressurized at 5 PSI utilizing a fixture which replaces the shorting pin. A leak rate in excess of 3.0 cubic centimeters per second shall be a reject.

4.4,14 Launcher environmental tests. When launchers are being procured less intervalometers the simulated firing test shall be deleted from the tests below. An intervalometer shall be installed for the Static Firing and Flight Firing Tests, any timing failure which can be attributed to the intervalometer shall be disregarded.

4.4.14.1 Initial examination. Prior to starting the environmental conditioning the following tests shall be performed. Visual examination for defects, pressure retention, simulated firing, insulation resistance, continuity, detent operation and loading.

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4,4.14.2 High temperature test. The launcher assembly without shipping covers shall be subjected to high temperature as specified in MIL-STD-810, Method 501, Procedure I. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects and the following tests shall be performed: Simulated firing, insulation resistance and continuity.

4.4.14.3 Low temperature test. The launcher assembly without shipping covers shall be subjected to low temperature of -54°C (-65°F) for 24 hours as specified in MIL-STD-810, Method 502. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects and the following tests shall be performed: Simulated firing, insulation resistance and continuity.

4.4.14.4 Shock test. The launcher assembly without shipping covers, shall be mounted on a bomb rack, loaded with seven (7) inert 2.75 Inch FFAR and subjected to shock in the vertical and longitudinal axis in accordance with MIL-STD-810, Method 516, Procedure I. The vertical half sine shock pulses shall have a peak of 10.0 g’s and an 18 ms duration, and the longitudinal half sine shock pulses shall have a peak of 13.5 g's and an 18 ms duration. The applied shock shall be measured midway between the attachment lugs on the bomb rack. The inert rockets shall not be released during the shock test. This test shall be performed with the intervalometer in the arm position with no power applied. Upon completion of the above, the sample shall be visually examined for damage or defects, and intervalometer advancement. The following tests shall be performed: Simulated firing, insulation resistance, continuity, detent operations and loading.

4.4.14.5 Vibration. The launcher loaded with seven (7) inert 2.75 Inch FFAR shall be subjected to vibration as specified in MIL-STD-810, Method 514, Procedure I, Curve C. Electrical contact between the launcher and each rocket motor shall be monitored continuously during the vibration, The following tests will be performed during a resonant dwell and at the end of vibration in each axis: Simulated Firing examination for intervalometer advancement. The inert rocket shall not be released during the vibration test. Immediately upon completion of the environmental conditioning, the following tests shall be performed: Visual examination for damage or defects and for intervalometer advancement, Simulated Firing Test, Continuity, Insulation Resistance and Detent Operation.

4.4.16.6 Rough handling. The launcher shall be fully loaded with 2.75 Inch FFAR Rockets and shipping covers in place. The launcher shall be dropped four times onto a solid concrete surface, falling freely through a vertical distance of 30 ± 1 inch and striking at an angle of 45 degrees ±1 degree. The 30 inch distance shall be measured from the lowest point on the launcher, when in the 45 degree attitude, to the solid concrete surface. The forward end of the launcher is to strike the concrete on two of the drops, and the aft end on the other two drops. Subsequently,

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the launcher must remain sealed. Immediately upon completion of the environmental conditioning, the following tests shall be performed: Pressure retention, detent operation and rocket loading. There shall also be a visual examination for physical damage to the launcher, and for intervalometer advancement.

4.4.14.7 Salt fog. The launcher with shipping covers in place shall be subjected to a salt fog test as specified in MIL-STD-810, Method 509 with the exception that the test period shall be 50 hours. This exposure shall not have deteriorated the operability of the launcher. Immediately upon completion of the environmental conditioning, the sample shall be visually examined for damage or defects and the following tests shall be performed: Simulated Firing, Insulation Resistance and Continuity. Upon completion of a 48 hour dry period, the above tests shall be repeated.

4.4.14.8 Static loading tests. The launcher without shipping covers shall be suspended from a pylon simulating or consisting of an Aero 14 type combination bomb rack and rocket launcher rack. The launcher shall be subjected to the following static loads in accordance with MIL-A-8591:

a. 2750 ± 20 pounds (13 g) acting downward

b. 1800 ± 20 pounds (8.5 g) to the side (horizontal)

c. 2750 ± 20 pounds (13 g) acting rearward

d. 2750 ± 20 pounds (13 g) acting forward

Application of test loading shall be achieved in such a manner that the loads are equally divided among the launching tubes and are uniformly distributed over the length of each. The side (horizontal) and downward loadings shall be applied simultaneously. The axes of the applied loads shall intersect on the centerline of the launcher package at a point midway between suspension hangers. The forward and rearward forces shall be applied along the centerline of the launcher package. The load shall be evenly distributed over the face of the aft bulkhead and the reaction of this load shall be evenly distributed through both hanger lugs. Test samples shall withstand the specified loads without structural failure. Upon completion of the environmental conditioning, the sample shall be visually examined for damaged or defects and the rocket loading test shall be performed.

4.4.14.9 Static firing test. The loaded launcher with shipping covers removed shall be suspended from a bomb rack or simulated bomb rack having 14 inch suspension provisions and test fired. The potential at the launcher connector shall be 20.0 ± 0.5 volts DC. There shall be 5 ohm limiting resistor in series with the launcher intervalometer. The launcher shall be capable of igniting all functional rockets in a normal manner, remain intact on the suspension device and sustain no damage other than peeling and/or darkening of the exposed surfaces. The firing sequence shall consist of ripple firing the loaded launcher followed by 23 rounds fired from one of the tubes and ripple firing

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the loaded launcher again. Data recorded during the ripple firings shall include time to the first firing pulse and intervals between pulses thereafter. Upon completion of the firings, the sample shall be visually examined for damage or defects and the following tests shall be performed: Detent operation, insulation resistance, continuity and the simulated firing test.

4.4.14.10 Flight firing. The launcher shall be loaded with 2.75 Inch FFAR and with fairings installed, if required by the service, shall be installed on a bomb rack having 14 inch suspension and flown to determine conformance with the requirement; and demonstrate ability to withstand air loads and vibration imposed by the aircraft for speeds UP to math 1.1. Prior to the flight test, the launcher shall be inspected to determine fit and comparability with the aircraft. The fully loaded launcher shall then be flown, fired in ripple mode, reloaded, flown and fired in the single mode. There shall be no damage to the aircraft, from fairing disintegration or launcher debris, which may-require replacement of aircraft skin.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with the contract.

5.2 Packing. Packing shall be in accordance with the contract.

5.3 Marking. Marking shall be in accordance with the contract.

6. NOTES

6.1 Intended use. The rocket launcher packages covered by this specification are intended for use as domestic and overseas shipping containers for 2.75 Inch Folding Fin Aircraft Rockets, and for service launching of 2.75 Inch Folding Fin Aircraft Rockets.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When the first article test samples and acceptance samples should be sent, the activity responsible for testing, and instructions concerning the submittal of the test reports (see 4.2).

c. Lot size maximum of one month’s production, plus lot acceptance samples. (see 4.3.1 and 4.3.2).

d. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

e. Requirements for preservation, packaging, packing and marking.

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6.3 Intervalometer. The requirements for the intervalometer are specified in Missile Interim Specification MIS-23156 Intervalometer, 2.75 Inch Rocket Launcher.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 1340-A319

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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OFFICIAL BUSINESS

Commanding General
U. S. Army Missile Command
ATTN: AMSMI-RCS
Redstone Arsenal, Alabama 35809

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